

CULDROSE, United Kingdom

WMO: 038090

Lat: 50.0844N

Lon: 5.2572W

Elev: 81

StdP: 100.35

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	0.3	-4.3	2.7	0.7	-2.8	3.0	2.0	16.0	8.6	14.6	8.0	4.6	50	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	21.6	17.4	20.2	16.6	19.1	16.2	18.3	20.4	17.5	19.2	16.9	18.3	4.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.7	18.9	16.9	12.2	18.2	16.3	11.7	17.7	52.0	20.7	49.6	19.2	47.8	18.3	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	12.1	10.9	-3.3	24.1	1.5	1.8	-4.4	25.4	-5.2	26.4	-6.1	27.4	-7.2	28.7		
(5)			-4.1	19.7	1.5	1.4	-5.2	20.7	-6.0	21.6	-6.9	22.4	-8.0	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	7.0	6.8	7.5	8.9	11.4	14.0	15.9	16.0	14.4	12.3	9.5	7.7		
	DBStd	4.00	2.79	2.54	2.22	2.09	2.03	1.85	1.73	1.54	1.67	2.14	2.45	2.89		
	HDD10.0	439	95	92	79	43	9	0	0	0	0	6	38	78		
	HDD18.3	2692	351	324	335	282	214	132	79	76	118	188	266	328		
	CDD10.0	796	2	1	2	11	53	119	182	186	133	77	22	8		
	CDD18.3	6	0	0	0	0	0	1	3	2	0	0	0	0		
	CDH23.3	11	0	0	0	0	0	1	6	3	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.5	6.6	6.2	6.0	5.4	5.3	4.9	4.6	4.6	5.1	5.7	5.8	6.5		
(15) Precipitation	PrecAvg	993	115	89	64	67	58	60	69	74	67	98	123	109		
	PrecMax	1257	195	191	145	163	140	136	210	197	177	163	249	228		
	PrecMin	790	27	17	20	11	8	8	8	21	6	33	41	33		
	PrecStd	135	48	48	31	39	31	38	43	45	39	35	49	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	12.1	13.5	17.0	19.9	22.5	24.1	23.3	20.7	17.7	14.6	13.3		
		MCWB	11.2	10.7	10.6	12.9	14.9	17.3	18.6	19.5	17.4	15.7	13.7	12.3		
	2%	DB	11.7	11.2	11.8	14.5	17.7	20.4	22.0	20.7	19.0	16.4	13.9	12.7		
		MCWB	10.9	10.1	10.0	11.4	14.0	15.8	17.5	17.6	16.4	14.8	13.0	11.9		
	5%	DB	11.1	10.6	11.1	13.0	15.9	18.6	20.5	19.6	18.0	15.7	13.4	12.2		
		MCWB	10.4	9.7	9.7	10.6	13.1	15.2	16.7	16.8	15.7	14.4	12.5	11.4		
	10%	DB	10.7	10.1	10.5	12.0	14.6	17.3	19.1	18.7	17.2	15.2	12.9	11.6		
		MCWB	9.9	9.2	9.3	10.0	12.3	14.5	16.2	16.4	15.3	14.0	12.1	10.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.2	11.5	13.4	16.1	18.3	19.2	19.8	18.1	16.2	14.2	12.7		
		MCDB	11.9	11.5	12.4	15.8	18.6	21.7	22.5	22.2	19.5	17.0	14.4	12.9		
	2%	WB	11.2	10.6	10.8	12.0	14.6	16.6	18.2	18.5	17.2	15.6	13.4	12.3		
		MCDB	11.4	10.9	11.3	13.6	16.8	19.1	20.9	20.0	18.2	16.0	13.7	12.5		
	5%	WB	10.7	10.1	10.2	11.2	13.6	15.7	17.4	17.6	16.6	15.0	12.9	11.8		
		MCDB	11.0	10.4	10.8	12.4	15.3	17.7	19.4	18.8	17.5	15.4	13.2	12.1		
	10%	WB	10.1	9.5	9.8	10.6	12.8	15.0	16.7	17.0	16.0	14.5	12.2	11.1		
		MCDB	10.6	10.0	10.3	11.6	14.1	16.7	18.3	18.1	16.8	15.0	12.8	11.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.5	4.8	5.6	5.4	5.4	5.3	5.3	5.4	4.6	4.3	4.3		
		MCDBR	3.7	4.0	5.0	6.8	7.3	7.8	7.7	6.4	6.1	4.4	3.6	3.5		
	5% WB	MCWBR	4.0	4.0	4.2	4.8	4.7	4.9	4.6	4.2	4.3	3.7	3.7	4.1		
		MCWBR	3.6	3.8	4.3	5.5	6.3	6.5	6.2	5.5	5.1	3.9	3.2	3.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.338	0.351	0.379	0.384	0.391	0.377	0.383	0.367	0.364	0.359	0.336	0.330		
		taud	2.356	2.378	2.323	2.341	2.364	2.434	2.427	2.486	2.478	2.449	2.404	2.321		
	Ebn at Noon	Ebn at Noon	686	781	821	860	867	881	870	869	832	763	686	630		
		Edh at Noon	68	84	105	114	116	109	109	98	89	77	64	61		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.83	1.63	2.78	4.51	5.44	5.96	5.49	4.72	3.59	2.05	1.07	0.67		
	RadStd	RadStd	0.07	0.16	0.31	0.30	0.40	0.56	0.36	0.33	0.24	0.13	0.08	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page